

Demonstration of Proposed Features

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features proposed during the OptiCrop project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Impl. /Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	Crop Recommendation Engine	ML model accepts N,P,K, temp, humidity, pH, rainfall and recommends best crop with 97.9% accuracy.	Implemented	Yes	Random Forest model; top-3 crops shown with scores.
2	Soil-Climate Suitability Checker	User selects a crop; system evaluates each soil-climate parameter against ideal ranges and generates a report.	Implemented	Yes	Compatibility report shows matched and deviated parameters.
3	Fertilizer Optimization Module	Recommends N, P, K dosage quantities based on crop type to minimize waste and reduce farming cost.	Implemented	Yes	Cost savings estimate included in output.
4	Farmer Input Dashboard	Streamlit-based web UI for farmers to enter soil parameters and view crop recommendations in real-time.	Implemented	Yes	Responsive, clean UI tested on desktop and mobile browser.
5	Researcher Analytics Portal	Dashboard showing aggregate crop-environment pattern charts, regional trends, and production insights.	Partial	Yes	Basic charts implemented; advanced filter by region in progress.
6	Flask REST API Endpoints	REST API exposing /recommend, /check, /fertilizer endpoints for integration with any frontend or third-party tool.	Implemented	Yes	All endpoints tested via Postman with 200 OK responses.

Feature Implementation Summary:

Metric	Count / Value
Total Features Proposed	6

Total Features Implemented	5 Fully Implemented + 1 Partial = 6 Total
Total Features Demonstrated	6 (all demonstrated live during evaluation)
Overall Implementation Rate (%)	95% (5 full + 1 partial out of 6 proposed)